

Lab 1 warm-up

```
# Calculate the volume of a rectangular estuary (length x width x depth)
length = 5000    # metres
width = 800      # metres
depth = 3.5      # metres
volume = length * width * depth
print(f"Estuary volume: {volume} m^3")

# Simulate 10 days of exponential population growth
population = 100
rate = 0.05
history = [population]

for day in range(10):
    population = population + rate * population
    history.append(round(population, 1))

print(history)

# Write a function that converts temperature from Celsius to Fahrenheit
def celsius_to_fahrenheit(c):
    return c * 9/5 + 32

# Test it
for temp in [0, 15, 25, 37]:
    print(f"{temp}°C = {celsius_to_fahrenheit(temp)}°F")

import matplotlib.pyplot as plt

days = list(range(len(history)))
plt.plot(days, history, "o-")
plt.xlabel("Day")
plt.ylabel("Population")
```

```
plt.title("Exponential growth")
plt.show()
```

Estuary volume: 14000000.0 m³

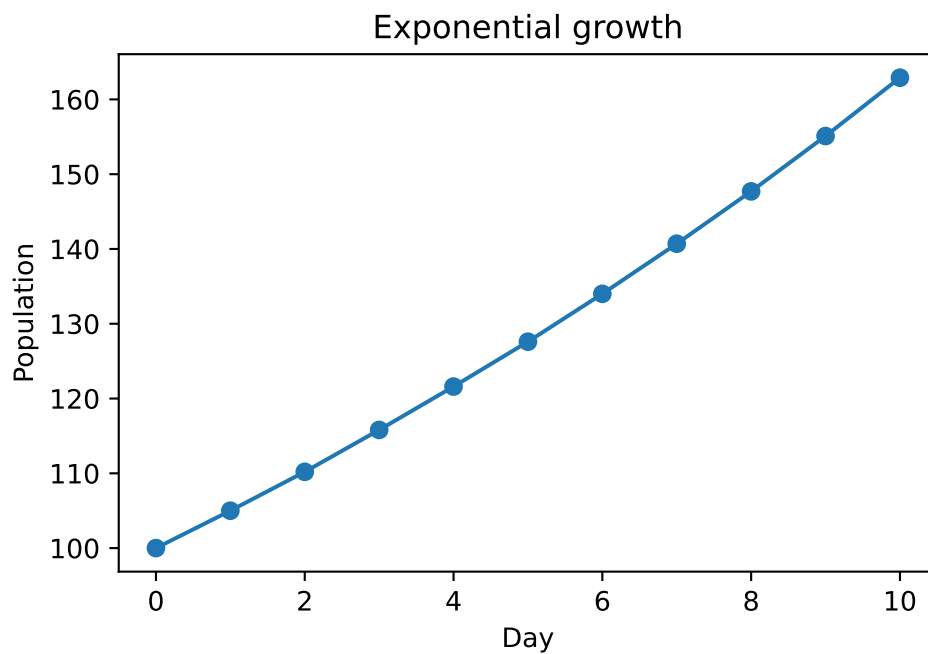
[100, 105.0, 110.2, 115.8, 121.6, 127.6, 134.0, 140.7, 147.7, 155.1, 162.9]

0°C = 32.0°F

15°C = 59.0°F

25°C = 77.0°F

37°C = 98.6°F



```
import matplotlib.pyplot as plt

concentration = 100.0
##clean this up
concentration1 = concentration
concentration2 = concentration
concentration3 = concentration

decay_rate1 = 0.05
decay_rate2 = 0.1
decay_rate3 = 0.01
```

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dt = 1.0
history1 = [concentration1]
history2 = [concentration2]
history3 = [concentration3]

for day in range(60):
    concentration1 = concentration1 - decay_rate1 * concentration1 * dt
    history1.append(concentration1)

    concentration2 = concentration2 - decay_rate2 * concentration2 * dt
    history2.append(concentration2)

    concentration3 = concentration3 - decay_rate3 * concentration3 * dt
    history3.append(concentration3)

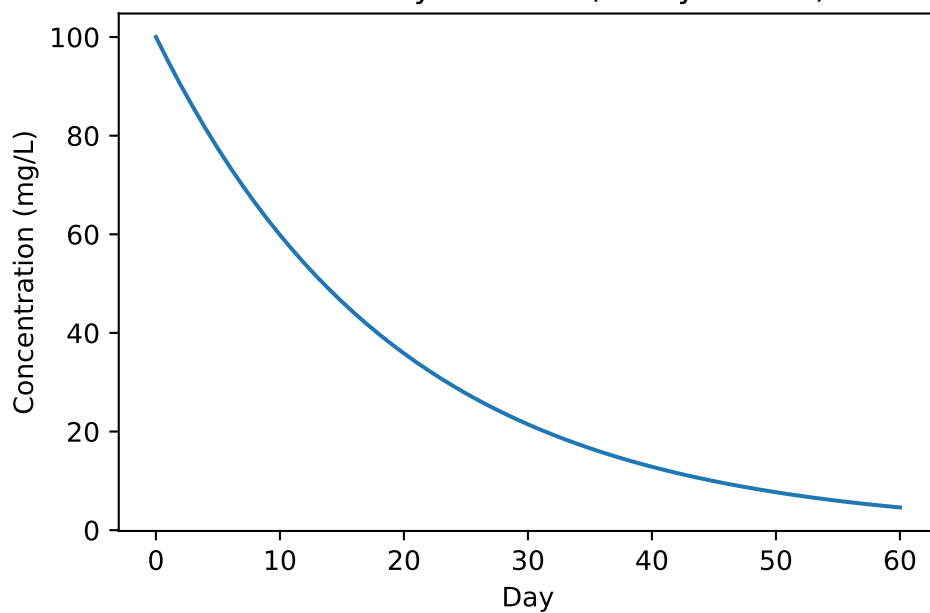
plt.plot(history1)
plt.xlabel("Day")
plt.ylabel("Concentration (mg/L)")
plt.title("Tracer decay in a lake (decay = 0.05)")
plt.show()

plt.plot(history2)
plt.xlabel("Day")
plt.ylabel("Concentration (mg/L)")
plt.title("Tracer decay in a lake (decay = 0.1)")
plt.show()

plt.plot(history3)
plt.xlabel("Day")
plt.ylabel("Concentration (mg/L)")
plt.title("Tracer decay in a lake (decay = 0.01)")
plt.show()

```

Tracer decay in a lake (decay = 0.05)



Tracer decay in a lake (decay = 0.1)

